Signed Retarded-Force Certificates for the Relativistic Vlasov–Maxwell System

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Abstract

The assigned manuscript claims unrestricted large-data classical regularity for the three-dimensional one-species relativistic Vlasov–Maxwell system. Its signed retarded-force identity already retains source acceleration, receiver differentiation, moving cutoffs, matrix weights and cone-entry terms. We preserve that identity rather than credit it as a new result. The local addition is an exact off-geometry defect formula, a tangential representation with its physical anisotropic error norm, and a weak impulse certificate for piecewise primitive approximations including their jumps. These results specify which approximation errors must be controlled and why ordinary small trajectory errors are insufficient near the cone. The weak certificate uses the original source-label measure; an approximate retarded geometry does not inherit the exact time-change Jacobian. Original Ram weak-adjoint and differentiated-jet arguments provide the comparison, with boundary and coefficient-derivative terms retained. Occupation estimates, angular summation, selection, bootstrap uniformity and continuation remain genuine upstream obligations. Contribution assessment: C2/5, the contribution is a modest source-specific defect and weak-error certificate, not a new cancellation identity or an independent proof of global Vlasov–Maxwell regularity.

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1 Model, claim and scope

Put $q(v) = \sqrt{1 + |v|^2}$, $u(v) = v/q(v)$. The source normalization is

$$\partial_t f + u \cdot \nabla_x f + (E + u \times B) \cdot \nabla_v f = 0, \tag{1}$$

$$\partial_t E - \nabla \times B = -j, \qquad \partial_t B + \nabla \times E = 0, \tag{2}$$

$$\nabla \cdot E = \rho, \quad \nabla \cdot B = 0, \qquad (\rho, j) = \int_{\mathbb{R}^3} (1, u) f \, \mathrm{d}v. \tag{3}$$

There is no neutralizing background and no 4π in (ρ, j). The data are $0 \le f_0 \in C_c^\infty(\mathbb{R}^6)$ and $E_0, B_0 \in C_b^\infty(\mathbb{R}^3) \cap L^2(\mathbb{R}^3)$, satisfying the two Gauss constraints. Here C_b^∞ means that each derivative is bounded, not that all derivatives have one common bound. The fields need not have all derivatives in L^2. Nonzero total charge and its Coulomb tail are allowed.

The source [1] asserts a unique classical solution, smooth on every finite interval, with compact particle phase support there, without size or symmetry restrictions. This report works on a compact interval of an *already existing* classical solution. All local vector norms are the physical Euclidean norms in $\mathbb{R}^3$, and matrix norms are induced operator norms. Nothing here constructs a global solution by a finite calibration or sampling argument.

The preserved conserved quantity is

$$H_0 = \frac{1}{2} \int_{\mathbb{R}^3} (|E|^2 + |B|^2) \, \mathrm{d}x + \iint_{\mathbb{R}^6} q f \, \mathrm{d}x \, \mathrm{d}v.$$

Particles obey $Y' = u(V)$, $V' = E + u(V) \times B$. Phase flow preserves six-dimensional volume and $f = f_0$ along this flow. Field energy alone cannot be substituted for the full energy in the source's occupation estimates.

1.1 Source architecture and the unreplaced interfaces

Source module	Mathematical role and boundary of this report
`setup` and introduction	Signed impulse bootstrap, dyadic momentum scale, and the harmonic-series obstruction to infinitely many momentum doublings in finite time.
`retarded`	Cone energy, source-label Jacobian, retarded potentials and initial-cone fields. Exact geometric identities are retained below.
`occupation`	Stopping cells, relative-velocity occupation and a joint energy budget. These are continuum estimates, not consequences of our finite fixtures.
`direct`	Nonnegative transport/source estimates, shared angular field budgets and the absolute-force estimate with a $\sqrt{w}$ loss.
`cancellation`	Signed source-trajectory integration by parts, mixed selected/unselected cutoffs, receiver coefficient and weighted boundary identity.
`direction`	First-direction exits, endpoint ramps and improved occupation on a restricted momentum/angle window.

Source module	Mathematical role and boundary of this report
`selection`	Fixed dyadic index selection, exponent contradiction, cutoff-neighbor summation and the signed weighted impulse estimate.
`closure`	Choice of constants in a strict order and removal of endpoint ramps by signed increments, not by absolute variation.
`continuation`	Finite-propagation modification of non-Sobolev field tails, local smooth construction and the imported bounded-momentum continuation criterion.

Every listed source module was literally read. Reading this dependency chain does not independently certify all its uniform estimates. The new propositions below do not replace any occupation, selection or continuation theorem.

2 The exact retarded law and its signed primitive

Fix a source label $z = (y_0, v_0)$ and receiver trajectory $X(t)$. On a noncollision retarded branch,

$$t = s + r, \quad r = |X(t) - Y(s,z)| > 0, \quad n = \frac{X(t) - Y(s,z)}{r}, \quad u = u(V(s,z)), \quad a = X_t(t).$$

Write

$$d = 1 - n \cdot u, \quad D = 1 - n \cdot a, \quad e = 1 - a \cdot u, \quad H = I + \frac{n \otimes a}{D}, \quad k = u - \frac{e}{D} n, \quad \mathcal{P} = \frac{k}{rd}. \tag{4}$$

Both d, D are positive because $|u|, |a| < 1$. A prime below is a derivative with respect to *source time* s; a_t is a derivative with respect to receiver time. Thus

$$\frac{\mathrm{d}t}{\mathrm{d}s} = \frac{d}{D}, \qquad r' = \frac{d}{D} - 1, \qquad rn' = N := \frac{d}{D}(a - n) + n - u, \qquad a' = \frac{d}{D} a_t. \tag{5}$$

In particular, $n \cdot N = 0$. Replacing a' by a_t is not an innocent notation change.

The source's combined transport and acceleration kernel, in source-label/source-time measure and suppressing the common $1/(4\pi)$, is

$$\mathcal{K} = -\frac{H}{r}\, \partial_u \left(\frac{u - n}{d} \right) [u'] - \frac{k(1 - |u|^2)}{r^2 d^2}. \tag{6}$$

The derivative ∂_u holds n fixed. The full trajectory derivative of $\mathcal{P}$ does not hold n, r, a fixed.

Proposition 2.1 (Source cancellation, retained rather than new)**.** *Under (5), the source identity is*

$$\mathcal{K} = -\mathcal{P}' + \mathcal{R} + \mathcal{A}, \tag{7}$$

$$\mathcal{R} = \frac{e}{r^2 D^2} \left(\frac{1 - |a|^2}{D} n - a \right), \qquad \mathcal{A} = \frac{n(a_t \cdot k)}{r D^2}. \tag{8}$$

Proof. Let $g = (u - n)/d$; then $Hg = k/d$. The u' terms in $\mathcal{K} + \mathcal{P}'$ cancel because $\partial_u (k/d) = H \partial_u g$. For the other terms use

$$\partial_a k[b] = \frac{n(b \cdot k)}{D}, \qquad N \cdot u - d r' - (1 - |u|^2) = -\frac{ed}{D}.$$

The n' part of $rd\,k'$ is

$$-\frac{ed}{D} \left(N + \frac{n(N \cdot a)}{D} \right).$$

Finally

$$N + \frac{n(N\cdot a)}{D} + k = \frac{d}{D}\left(a - \frac{1-|a|^2}{D}n\right).$$

Substitution gives exactly (8). These are the algebraic steps of the source's `can:identity`, reproduced to fix signs and clocks. □

2.1 Physical measure, weights and the initial cone

For the actual phase flow the retarded change of variables is

$$f_0(z)\,\mathrm{d}z\,\mathrm{d}s = Df(s,y,v)\,\mathrm{d}t\,\mathrm{d}y\,\mathrm{d}v. \tag{9}$$

The underlying label Jacobian at fixed receiver event is $\mathrm{d}z = d\ \mathrm{d}y\,\mathrm{d}v$; combining it with $\mathrm{d}t/\,\mathrm{d}s = d/D$ gives (9). These assertions require the actual retarded root and phase flow. The source excludes a null set of collision labels and justifies the remaining integrations by a tip cutoff. Its compact-solution dominated bounds justify identities; they are not uniform momentum estimates.

Let ψ be a Lipschitz receiver-time weight vanishing at the endpoints of its interval, $\Pi(t)$ a bounded C^1 matrix, and χ the sum of a *fixed* selected set of smooth dyadic cutoffs. With $K_t = fD\mathcal{K}$, $R = fD\mathcal{R}$, $A = fD\mathcal{A}$, the source formula is

$$\int \psi\chi\Pi K_t = B_0 + \int \Big[\psi\chi\Pi(R+A) + \psi_t\chi\Pi\,\frac{fk}{r} + \psi\chi\Pi_t\,\frac{fk}{r} + \psi\Pi\,\frac{fDk}{rd}\,\chi_s'\Big]\,\mathrm{d}t\,\mathrm{d}y\,\mathrm{d}v. \tag{10}$$

The notation K_t denotes a density in receiver time, not a derivative. All three derivatives of the weight product are indispensable.

At a source-entry point $s = 0$, $r = t_0$, the contribution has *positive* sign:

$$B_0 = \int f_0(z)\,\psi(t_0)\chi(0,z)\Pi(t_0)\frac{k_0}{t_0 d_0}\,\mathrm{d}z, \tag{11}$$

where the integral is over the labels entering in the receiver interval. This B_0 arises in the signed integration by parts. The initial-cone field in the retarded field representation and the homogeneous wave terms must also be included when reconstructing the full Lorentz force. They are not removed by naming (11) a boundary correction. The entry parametrization $y_0 = X(t_0) - t_0 n$ has Jacobian $t_0^2 D$. Initial bounded momentum and $Dk = Du - en$ then give the source estimate $|B_0| \le C_{f_0}\int|\psi|\ \mathrm{d}t_0$.

The source already differentiates a selected cutoff by the complementary unselected partition after factoring its *common physical multiplier*. It does not estimate a bump derivative by that same bump, nor differentiate a time-varying discrete selection. Our later weak error estimate does not remove these requirements.

3 An exact off-geometry defect formula

The next result is local algebra for approximate data, not an alternative retarded measure formula. Let r, n, u, a be C^1 functions of a parameter s, with $r > 0, |n| = 1, |u|, |a| < 1$. Let $b(s)$ be a supplied receiver-time derivative and $\zeta(s)$ a supplied inverse energy square. Define $d, D, e, H, k, \mathcal{P}$ by (4) and set

$$\epsilon_r = r' - (d/D - 1), \qquad \epsilon_n = rn' - N, \qquad \epsilon_a = a' - (d/D)b, \qquad \epsilon_q = \zeta - (1-|u|^2). \tag{12}$$

Here $\epsilon_n \perp n$ exactly; no independent normal component of n' is permitted. Put $\mathcal{K}_\zeta = -H\partial_u g[u']/r - k\zeta/(r^2 d^2)$ and let $\mathcal{R}, \mathcal{A}_b$ be (8) with $a_t = b$.

Theorem 3.1 (Four residuals with physical denominators)**.** *For these data,* $\mathcal{K}_\zeta = -\mathcal{P}' + \mathcal{R} + \mathcal{A}_b + \mathcal{Z}$*, where*

$$\begin{aligned}\mathcal{Z} = &-\frac{k\epsilon_r}{r^2 d} - \frac{e}{r^2 dD}\left(\epsilon_n + \frac{n(a\cdot\epsilon_n)}{D}\right) + \frac{k(u\cdot\epsilon_n)}{r^2 d^2}\\ &+ \frac{n(\epsilon_a\cdot k)}{rdD} - \frac{k\epsilon_q}{r^2 d^2}.\end{aligned} \tag{13}$$

This identity has no assumed smallness and holds in the given parameter and its specified measure.

Proof. The u' cancellation in Proposition 2.1 is unchanged. Relative to its kinematics, the increment in $\mathcal{P}'$ is

$$\partial_r \mathcal{P}\,\epsilon_r + \partial_n \mathcal{P}[\epsilon_n/r] + \partial_a \mathcal{P}[\epsilon_a].$$

Direct differentiation, with n-variations tangent, gives

$$\begin{aligned}\partial_r \mathcal{P}\,h &= -kh/(r^2 d),\\ \partial_n \mathcal{P}[v] &= -\frac{e}{rdD}\left(v + \frac{n(a\cdot v)}{D}\right) + \frac{k(u\cdot v)}{rd^2},\\ \partial_a \mathcal{P}[h] &= \frac{n(h\cdot k)}{rdD}.\end{aligned}$$

Replacing $1-|u|^2$ in the kernel by ζ adds $-k\epsilon_q/(r^2d^2)$. Summing gives (13), with all signs fixed by $\mathcal{K}_\zeta + \mathcal{P}' - (\mathcal{R} + \mathcal{A}_b)$. □

Corollary 3.2 (A nonasymptotic pointwise certificate)**.** *For the same data,*

$$\begin{aligned}|\mathcal{Z}| \le &\frac{|k|}{r^2 d}\,|\epsilon_r| + \frac{e}{r^2 dD}\left(|\epsilon_n| + \frac{|a\cdot\epsilon_n|}{D}\right) + \frac{|k|\;|u\cdot\epsilon_n|}{r^2 d^2}\\ &+ \frac{|\epsilon_a\cdot k|}{rdD} + \frac{|k|}{r^2 d^2}\,|\epsilon_q|\,.\end{aligned} \tag{14}$$

All factors are evaluated on the represented path. Correlated contractions can be retained instead of bounding every dot product by two norms.

Proof. Apply the Euclidean triangle inequality to (13). Since $|a|\,,|u|<1$, $e>0$; thus every displayed coefficient is nonnegative. □

In the source's small sector write $d = 2\theta^2, D = 2\phi^2$ and $\theta \le \kappa\phi$, for a fixed sufficiently small κ. The elementary source estimates $e \le C\phi^2$, $|k| \le C\theta/\phi$, and tangency give

$$|u\cdot\epsilon_n| \le C\theta\,|\epsilon_n|\,, \qquad |a\cdot\epsilon_n| \le C\phi\,|\epsilon_n|\,.$$

Consequently a convenient, deliberately nonoptimal bound is

$$|\mathcal{Z}| \le C_\kappa\left(\frac{|\epsilon_r|}{r^2\theta\phi} + \frac{|\epsilon_n|}{r^2\theta^2\phi} + \frac{|\epsilon_a|}{r\theta\phi^3} + \frac{|\epsilon_q|}{r^2\theta^3\phi}\right). \tag{15}$$

For example, the second term follows by bounding the two ϵ_n terms of (14) separately, using $\phi \le 1$. The last term records the particularly costly error of inserting an independently rounded mass-shell factor.

A necessary limitation. Off-geometry paths generally have neither the phase-space Jacobian nor the clock in (9). We therefore do not multiply (13) by fD and call it an error in the physical impulse. The safe use is in a fixed source-label integral with certified weights, or together with a separately proved measure/time-change error. If an approximate path and an exact path are compared, state enclosures, retarded-root comparison, initial-entry changes and law errors are additional premises. An exact formula for the approximate path alone is not such a comparison.

4 Separating source-angle and receiver-angle conditioning

For fixed r, n, a, put $u_\perp = u - (n \cdot u)n$, $a_\perp = a - (n \cdot a)n$, and

$$T(u) = \frac{u_\perp}{1 - n \cdot u} \in n^\perp.$$

This tangential variable is not the particle momentum or a change in the physical norm.

Proposition 4.1 (Exact primitive metric)**.** *For every $|u| < 1$,*

$$\mathcal{P}(u) = \frac{1}{r}\left(T(u) + \frac{n}{D}(a_\perp \cdot T(u) - 1)\right). \tag{16}$$

For two velocities $u, \widetilde{u}$, with the same r, n, a, let $v = T(u) - T(\widetilde{u})$. Then

$$|\mathcal{P}(u) - \mathcal{P}(\widetilde{u})|^2 = \frac{1}{r^2}\left(|v|^2 + \frac{(a_\perp \cdot v)^2}{D^2}\right). \tag{17}$$

The maximal amplification of tangent error is

$$\frac{1}{r}\sqrt{1 + \frac{|a_\perp|^2}{D^2}} \le \frac{1}{r}\sqrt{\frac{2}{D}}.$$

Proof. Writing $u = u_\perp + u_n n$, $a = a_\perp + a_n n$, one finds

$$u_n - \frac{e}{D} = \frac{u_n - 1 + a_\perp \cdot u_\perp}{D} = \frac{-d + a_\perp \cdot u_\perp}{D}.$$

Division by rd proves (16). The two components of its difference are orthogonal, proving (17). The tangent Gram operator is $I_{n^\perp} + a_\perp \otimes a_\perp / D^2$. Its largest eigenvalue is $1 + |a_\perp|^2 / D^2$. Finally $|a_\perp|^2 < 1 - a_n^2 = D(1 + a_n)$ and $D = 1 - a_n$. □

This exact metric retains the physical receiver projection and can be strictly better than replacing it by its largest eigenvalue. If $a_\perp = 0$, tangent errors are scaled just by $1/r$. For $\Delta u = u - \widetilde{u}$ there is also the exact identity

$$T(u) - T(\widetilde{u}) = \frac{(\Delta u)_\perp + T(\widetilde{u})(n \cdot \Delta u)}{d}. \tag{18}$$

Thus the receiver cost $D^{-1/2}$ is distinct from the source denominator d^{-1}. A relative enclosure $|n \cdot \Delta u| \le \gamma \widetilde{d}$, $0 \le \gamma < 1$, implies $d \ge (1 - \gamma)\widetilde{d}$ and makes (18) a usable interval bound. An absolute tolerance on Δu need not.

Example 4.2 (Vanishing velocity error and diverging primitive error)**.** Take $r = 1, n = (0, 0, 1), a = 0$, and $0 < h \le 1/2$. Let

$$u = (h, 0, 1 - 2h^2), \qquad \widetilde{u} = (h, 0, 1 - h^2).$$

Both have norm strictly less than one, $|u-\widetilde{u}|=h^2$, while

$$d=2h^2,\quad \widetilde{d}=h^2,\quad |\mathcal{P}(u)-\mathcal{P}(\widetilde{u})|=\frac{1}{2h}.$$

Here the receiver metric is perfectly conditioned: $D=1, a_\perp=0$. The failure is entirely in the source-angle map T. These are admissible instantaneous velocities, not a constructed Vlasov–Maxwell solution or a counterexample to the source theorem.

5 A weak certificate with entry and panel jumps

The following statement uses a fixed parameter and a fixed measure. It may be applied to actual source branches with label measure $f_0(z)\,\mathrm{d}z/(4\pi)$. It may also be applied to a specified approximate branch, but then certifies only that branch's functional.

Theorem 5.1 (Piecewise primitive error, without differentiating the error)**.** *Let $[\alpha,\beta]$ be compact, $\mathcal{P}\in W^{1,1}([\alpha,\beta];\mathbb{R}^3)$, $C\in L^1([\alpha,\beta];\mathbb{R}^3)$, and $W\in W^{1,\infty}([\alpha,\beta];\mathbb{R}^{m\times 3})$. Define*

$$I=\int_\alpha^\beta W(-\mathcal{P}'+C)\,\mathrm{d}s.$$

On a finite partition $\alpha=s_0<\cdots<s_N=\beta$, let $\widetilde{P}$ be piecewise $W^{1,1}$, with one-sided traces, and let $\widetilde{C}\in L^1$. Define the approximating functional by

$$\widetilde{I}=-W(\beta)\widetilde{P}(\beta-)+W(\alpha)\widetilde{P}(\alpha+)+\sum_j\int_{s_{j-1}}^{s_j}(W'\widetilde{P}+W\widetilde{C})\,\mathrm{d}s. \tag{19}$$

Suppose $\left|\mathcal{P}-\widetilde{P}\right|\le\varepsilon_P$ almost everywhere and $\left|C-\widetilde{C}\right|\le\varepsilon_C$, with endpoint errors $\varepsilon_\alpha,\varepsilon_\beta$. Then

$$\begin{aligned}\left|I-\widetilde{I}\right|\le{}&\|W(\alpha)\|\,\varepsilon_\alpha+\|W(\beta)\|\,\varepsilon_\beta\\&+\int_\alpha^\beta\|W'\|\,\varepsilon_P\,\mathrm{d}s+\int_\alpha^\beta\|W\|\,\varepsilon_C\,\mathrm{d}s.\end{aligned} \tag{20}$$

Equivalently, the functional (19) is

$$\sum_j\int_{s_{j-1}}^{s_j}W(-\widetilde{P}'_{\mathrm{ac}}+\widetilde{C})\,\mathrm{d}s-\sum_{j=1}^{N-1}W(s_j)[\widetilde{P}]_j,\qquad [\widetilde{P}]_j=\widetilde{P}(s_j+)-\widetilde{P}(s_j-). \tag{21}$$

Proof. Sobolev integration by parts for the exact primitive gives $I=-[W\mathcal{P}]_\alpha^\beta+\int(W'\mathcal{P}+WC)$. Subtract (19) and use the physical operator norm, keeping the matrix on the left of each vector. This proves (20). Integrating by parts on each open panel gives (21): the internal endpoint contributions to $\int -W\widetilde{P}'_{\mathrm{ac}}$ are $+W(s_j)[\widetilde{P}]_j$, which must be subtracted. No derivative of $\mathcal{P}-\widetilde{P}$ was estimated. □

Corollary 5.2 (Source-label impulse certificate)**.** *Let almost every label z have a branch $[\alpha(z),\beta(z)]$ satisfying Theorem 5.1, with $W=\psi(t(s,z))\chi(s,z)\Pi(t(s,z))$. Suppose the branch functions and the errors are measurable and the right-hand side of (20) is integrable against $\mathrm{d}\mu(z)=f_0(z)\,\mathrm{d}z/(4\pi)$, and that the exact branch functional $I(z)$ is μ-integrable. Then the error in the label-averaged impulse is at most the integral of that right-hand side against μ. For an exact branch take $C=\mathcal{R}+\mathcal{A}$; for an off-geometry branch take $C=\mathcal{R}+\mathcal{A}_b+\mathcal{Z}$.*

Proof. Integrate the branchwise inequality and use Tonelli for its nonnegative right-hand side and the triangle inequality for the vector integral. The integrability premise is essential, in particular near $r = 0$ and small d, D. □

The product derivative in this corollary is exactly

$$W' = \frac{d}{D}\,\psi_t \chi \Pi + \psi \chi'_s \Pi + \frac{d}{D}\,\psi \chi \Pi_t \tag{22}$$

on an actual branch. On an approximate branch its actual t', not an asserted d/D, must replace both clock factors. Thus the weak certificate does not forget any term of (10). Its advantage is that certified errors in the primitive need not be differentiated. Its cost is the weighted total variation of W, the entry error and the full label integral. Increasing the number of narrow panels or cutoff transitions can increase this cost even if the pointwise approximation improves.

5.1 What must be supplied for a finite implementation

On finitely many enclosed branch panels, a valid certificate consists of:

1. intervals for r, d, D bounded away from zero and bounds for the actual fields, trajectories and weights, not just point samples;
2. a certified ε_P, for example from (17)–(18), with separate variation bounds if r, n, a also change;
3. bounds for the bulk error, including $\mathcal{Z}$ when appropriate, and endpoint and jump traces;
4. rigorous integration enclosures for (19) and (20), in the same measure;
5. a bound for every omitted label, time interval, velocity region, cone-tip region and dyadic cutoff overlap.

If a quadrature error is bounded by E_{quad} and the omitted part of the *impulse* by E_{tail}, these two nonnegative numbers add to (20). A small omitted label probability alone is not an E_{tail}: the retarded integrand may be singular. One may instead supply an actual uniform integrand envelope, a weighted integrability estimate, or the source's analytical tail estimates with all their premises.

If the input law itself changes from μ to $\widetilde{\mu}$, and $J(z)$ is the exact branch functional, then a separate safe estimate is

$$\left| \int J \,\mathrm{d}\mu - \int J \,\mathrm{d}\widetilde{\mu} \right| \le \int |J| \;\mathrm{d}\,|\mu - \widetilde{\mu}|\,.$$

Using $\|J\|_\infty$ here requires that norm to be finite. Total variation of laws without integrand control is not sufficient near the cone. This separates finite integration errors from actual law and geometry errors.

5.2 Exact negative controls

For a scalar illustration of the same ordered integration by parts, take $W(s) = s^2, P(s) = s^3, C = 0$ on $[0, 1]$. Then

$$\int_0^1 -WP' \,\mathrm{d}s = -\frac{3}{5} = -1 + \int_0^1 W'P \,\mathrm{d}s = -1 + \frac{2}{5}.$$

Discarding the weight derivative gives -1, not $-3/5$. Taking $W = 1, P = 1$ on a branch shows that the two endpoint terms must cancel; retaining only a supposedly negligible lower endpoint is incorrect. In the actual source, the upper time weight can vanish while the entry weight does not, leaving precisely the positive entry term.

For a panel jump, take $W = 1$ and $\widetilde{P} = 0$ before $1/2$, 1 afterwards. Its ordinary panel derivative is zero, but the derivative as a measure is $\delta_{1/2}$; hence the correct contribution of $-\mathrm{d}\widetilde{P}$ is -1. The correction in (21) is not optional even when each panel polynomial is evaluated exactly.

Finally, Example 4.2 shows that a vanishing velocity error does not produce a vanishing primitive error without a cone margin. Neither bounded coefficient vectors in a finite fit nor a small unweighted residual controls these three failures.

6 Original Ram comparison and derivative accounting

The original TR207 weak construction [7] starts from an actual differential operator $Lf = \sum_{n=0}^{N} k_n f^{(n)}$, not from a free matrix model. With a compactly supported test function Θ, the bilinear weak pairing is

$$\int (Lf)\Theta = \int f \sum_{n=0}^{N} (-1)^n \partial^n (k_n \Theta).$$

For a test touching endpoints it instead includes

$$\sum_{n=1}^{N} \left[\sum_{j=0}^{n-1} (-1)^j f^{(n-1-j)} \partial^j (k_n \Theta) \right]_{\ell}^{r}.$$

There is no justification for dropping this sum merely because an atom is numerically small. The fresh original sections include the exact boundary formula, its triangle-inequality leakage bound, finite-class rank and perturbation statements. Finite-class invertibility is not invertibility on the continuum of source labels.

The freshly read original jet chapter [8] likewise retains

$$\partial^k \left(\sum_{n=0}^{N} S_n f^{(n)} \right) = \sum_{p=0}^{k} \binom{k}{p} \sum_{n=0}^{N} S_n^{(k-p)} f^{(n+p)}.$$

Rows discarded because $n + p > N$ contribute to the remainder unless f is actually in the stated finite polynomial class. Moving integration limits produce endpoint terms, and differentiated remainders need their own bounds. These are useful warnings against treating a finite derivative table as a full Vlasov identity.

The comparison here is precise: P is the physical retarded primitive, W is the actual product of time, cutoff and projection weights, and the weak derivative transfer is (19). The source already performs this transfer exactly. The added certificate allows a discontinuous panel representation of that same primitive and explicitly charges its error, jumps and entry. The off-geometry formula specifies the extra error before any such transfer. Classical integration by parts, differentiation and operator-norm inequalities are not claimed as new Ram theorems.

7 Why the global closure remains upstream

Let P now denote the source's dyadic upper momentum scale, rather than the primitive $\mathcal{P}$, and put $L = \log(2 + P)$, $S(w) = P^2 L / w$. The source bootstraps signed impulses on eligible receiver intervals:

$$|V_X(t_2) - V_X(t_1)| \le MP\sqrt{I} + AIS(w), \qquad I = t_2 - t_1. \tag{23}$$

Eligibility requires receiver energy comparable to w. Its nonnegative short-interval estimates may integrate over measurable subsets of eligible times. The final signed estimate is for the actual interval and cannot be obtained by inserting arbitrary signs into a nonnegative bound.

In `direct`, the absolute force estimate still has a $\sqrt{w}$ loss. In `selection`, the selected receiver coefficient is $O(w^{-1/2})$ after the weighted dyadic sum, which is designed to cancel this loss. The dimensionless weight

$$\mathcal{W} = (P\theta)^{-\varepsilon}\phi^{\varepsilon}h^{\varepsilon}, \qquad \varepsilon = 10^{-5},$$

and the improved occupation exponent 29/25 enter a specific exponent contradiction. A compact-grid check of (13) says nothing about that continuum contradiction, the selection complement or the omitted angular scales.

After endpoint ramps of relative length η are restored by signed increments, the source obtains the bound

$$\begin{aligned}(C_\eta + 2M\sqrt{\eta})P\sqrt{I} \\ + [C_\eta(M + \sqrt{A}) + 2A\eta]IS(w).\end{aligned} \tag{24}$$

Its order is to choose η first, then M, then A, and only then a large threshold P_0. Constants justified on one fixed compact solution interval cannot silently enter these supposedly uniform choices.

Proposition 7.1 (Conditional allowance for a certified added error)**.** *Suppose, with the source's quantifiers, that a complete approximation and tail analysis adds at most*

$$E_{\mathrm{add}} \le e_1 P\sqrt{I} + e_2 IS(w)$$

to (24)*, uniformly over every eligible receiver and interval. If for some* $0 < \vartheta < 1$

$$C_\eta + 2M\sqrt{\eta} + e_1 \le \vartheta M, \qquad C_\eta(M + \sqrt{A}) + 2A\eta + e_2 \le \vartheta A,$$

then that step retains a strict ϑ*-improvement of* (23)*.*

Proof. Add the certified error to (24) and compare its two nonnegative scale coefficients. No limit or sampling argument is involved. □

This elementary interface is useful only if *both* coefficients are proved uniformly. Theorems 3.1 and 5.1 do not supply that uniformity. A fixed absolute error can be larger than $P\sqrt{I}$ on sufficiently short intervals.

The source then turns uniform signed estimates into a lower bound of order $1/\log(2 + 1024\,2^j)$ for successive momentum-doubling times; their sum diverges. Its continuation module separately modifies field tails by a divergence-free compact construction plus a vacuum Maxwell field, retaining the Coulomb part required by Gauss' law. It invokes bounded-momentum continuation for Sobolev data and transfers back by finite propagation. This is not an assumption that the original $C_b^\infty \cap L^2$ fields already lie in every H^k. The Luk–Strain statement of the criterion [2], Theorem 1.1 and footnote 1, uses H^5 data and a bounded continuous momentum-support function. Its 4π-density convention is matched by $f_{\mathrm{LS}} = f/(4\pi)$, with fields unchanged.

8 Current literature, value and limitations

The source is the assigned September 23 manuscript in the publisher repository, not a replacement file fetched from a later search. Fresh public path-history metadata on October 9 returned the initial commit of October 6, `adc7f1241b42...`; this is a byte/history statement, not mathematical validation. The publisher's public formalization-scope metadata [6] claims a corresponding global result. No Lean code was executed or independently audited here.

The classical bounded-momentum continuation statement was freshly checked in the primary Luk–Strain text, including its regularity footnote and normalization. The full external proof was not audited. Current primary abstracts list Wang's cylindrical-symmetry sequence: Part I, version 2 of July 16, 2026, and Part II, version 1 of that date [3, 4]. These retain symmetry and are not an independent confirmation of the assigned nonsymmetric claim. Bigorgne's version 2 and published 2025 article [5] allow a large Maxwell field with a small distribution function. That is also a different hypothesis class. Only abstract/version/publication metadata was inspected for these comparisons, not their complete proofs.

The practical advantage of the local results is an error ledger whose terms have a physical meaning: cone distance, source angle, receiver angle, time conversion, mass shell, cutoff variation, initial entry, and panel jumps. The exact tangent metric can preserve correlations that a scalar condition number loses. The weak formulation avoids a derivative of the approximation error. These are modest refinements of an already careful source identity.

The costs are equally explicit. Small denominators remain; endpoint and tail estimates must be supplied; the source law cannot be changed silently; and narrower weights can increase variation. No practical speedup, computable global momentum bound, historical priority or new large-data theorem is established. No mathematical correction to the source cancellation formula was required in this scoped derivation.

Further work could develop interval enclosures for exact retarded roots, relative d, D margins and entry maps, followed by certified label and angular tails. A useful stronger result would prove that the resulting errors satisfy the two uniform scale budgets in Proposition 7.1. Until then, its hypotheses are a research target rather than an achieved continuum certificate.

9 Reading and reproducible checks

All 16 assigned public text files were freshly read in bounded literal displays: 14 TeX files, the bibliography and README, totaling 4,019 lines, with no unread public-text complement. This includes all ten scientific section modules and all three figure source files, not merely the driver and cancellation section. File hashes were persisted before the displays and matched after them. The local ledgers retain each successful interval.

The original TR207 file was freshly read at lines 127–364 and 386–505, totaling 358 lines. The original jet chapter was read at 500–676, 733–850, and 890–941, totaling 347 lines. Their unread complements are explicitly recorded; the rest of the confidential corpus was not read. These are original proof passages, not earlier comparative reports. The selected originals have 2,371 unread lines in total. Neither a catalogue reference nor a source bibliography citation counts as reading its target.

The final owned script performs 54 exact cases and 273 assertions: 24 instantaneous rational geometries for the source identity and full defect formula; 15 tangent-metric cases; five polynomial weak-impulse and jump cases; nine near-cone negative controls; and one additional endpoint/weight diagnostic. There are no floating diagnostics and no parent or foreign cases in these totals. The script differentiates the primitive independently before comparing it with the residual formula. Its rational normal vectors include non-axis directions, and its matrix weights vary with the parameter.

These checks validate finite algebraic fixtures only. They do not test global existence, uniform occupation, infinitely many dyadic bins, singular tails, all initial data, or the imported continuation theorem. Proofs, rather than the counts, establish the local propositions.

References

[1] OpenAI, *Global classical solutions of the three-dimensional relativistic Vlasov–Maxwell system*, September 23, 2026, assigned publisher source. Publisher source directory (public repository). Pinned local scientific text; global conclusions treated as source claims.

[2] J. Luk and R. M. Strain, *A new continuation criterion for the relativistic Vlasov–Maxwell system*, Communications in Mathematical Physics 331 (2014), 1005–1027. https://arxiv.org/abs/1406.0165. Fresh statement and footnote inspection in the primary HTML, lines 39–90.

[3] X. Wang, *Large data global solution of the 3D RVM system with cylindrical symmetry I: Iterative smoothing scheme*, arXiv:2203.01199v2, July 16, 2026. https://arxiv.org/abs/2203.01199.

[4] X. Wang, *Large data global solution of the 3D RVM system with cylindrical symmetry II: Pointwise estimates*, arXiv:2607.14685v1, July 16, 2026. https://arxiv.org/abs/2607.14685.

[5] L. Bigorgne, *Global existence and modified scattering for the solutions to the Vlasov–Maxwell system with a small distribution function*, Analysis & PDE 18 (2025), 629–714. https://arxiv.org/abs/2208.08360; version 2, August 31, 2023.

[6] OpenAI, publisher formalization-scope metadata, https://github.com/openai/math/blob/main/lean/docs/362.md. Public scope statement only; no independent execution or formal proof audit.

[7] M. Subbarao, original confidential TR207, *Ram-Transform robust calibration and inversion*, local file `TR207-rmt-robust-calibration-inversion.tex`. Fresh proof scope: lines 127–364 and 386–505. Boundary, exact weak-adjoint and finite-class perturbation arguments; not a Vlasov–Maxwell theorem.

[8] M. Subbarao, original confidential nonlinear integral/PDE jet chapter, local `TR1-lin-nonlin-rev2/sec-03-jets.tex`. Fresh scope: lines 500–676, 733–850, 890–941. Leibniz rows, moving endpoints, differentiated remainders and finite-class recovery with stated hypotheses.